

Electromagnetic Radiation

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Chapter Overview

This chapter examines electromagnetic radiation—the generation and propagation of waves from time-varying sources. We begin by introducing the scalar and vector potentials used to compute radiated fields, followed by the application of Green functions to solve the inhomogeneous Helmholtz equations governing these potentials. The general free-space solution and the far-field approximation are then derived, together with the Stratton–Chu formulation. These tools are applied to the Hertzian dipole to analyze its radiation properties and field patterns, after which image theory is introduced. The chapter concludes with the analysis of radiation from an aperture.

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Vector and Scalar Potentials

In the absence of magnetic sources, Gauss' law for magnetism states that $\vec{\nabla} \cdot \vec{B} = 0$. A key vector identity states that the divergence of the curl of any vector field is always zero. Conversely, on the topologically simple free-space domain used here (or locally on a contractible subregion), a smooth divergence-free field can be expressed as a curl. Thus the magnetic flux density $\vec{B}$ may be written in terms of a magnetic vector potential $\vec{A}$:

$$\vec{B} = \vec{\nabla} \times \vec{A}. \quad (3.1.1)$$

Substituting this into Faraday's law (1.4.5) with no magnetic sources ($\vec{M} = 0$), we obtain:

$$\vec{\nabla} \times (\vec{E} + i\omega\vec{A}) = 0. \quad (3.1.2)$$

In a simply connected region, a curl-free vector is the gradient of a scalar function. Choosing that gradient to be $-\vec{\nabla}\varphi$ defines

$$\vec{E} = -\vec{\nabla}\varphi - i\omega\vec{A}. \quad (3.1.3)$$

Vector and Scalar Potentials (cont.)

In a simple medium, substituting (3.1.1) and (3.1.3) into Ampère's law (1.4.6) gives

$$\begin{aligned}\frac{1}{\mu} \vec{\nabla} \times \vec{\nabla} \times \vec{A} &= \vec{\nabla} \times \vec{H} = i\omega \vec{D} + \vec{J} \\ &= i\omega\epsilon \left(-\vec{\nabla}\varphi - i\omega\vec{A} \right) + \vec{J} \\ &= -i\omega\epsilon \vec{\nabla}\varphi + \omega^2\epsilon\vec{A} + \vec{J}.\end{aligned}$$

Multiplying by μ gives the wave-like equation for $\vec{A}$:

$$\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \mu\vec{J} + \omega^2\mu\epsilon\vec{A} - i\omega\mu\epsilon\vec{\nabla}\varphi. \quad (3.1.4)$$

Substituting (3.1.3) into Gauss' law (1.4.7), with constant scalar permittivity, gives every intermediate term:

$$\begin{aligned}\rho &= \vec{\nabla} \cdot \vec{D} = \epsilon \vec{\nabla} \cdot \vec{E} \\ &= \epsilon \vec{\nabla} \cdot \left(-\vec{\nabla}\varphi - i\omega\vec{A} \right) \\ &= -\epsilon \nabla^2 \varphi - i\omega\epsilon \vec{\nabla} \cdot \vec{A}.\end{aligned}$$

Vector and Scalar Potentials (cont.)

Dividing by $-\epsilon$ yields the scalar-potential equation

$$\nabla^2 \varphi + i\omega \vec{\nabla} \cdot \vec{A} = -\frac{\rho}{\epsilon}. \quad (3.1.5)$$

Using the vector identity $\vec{\nabla} \times \vec{\nabla} \times \vec{A} = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$, and rearranging terms, equations (3.1.4) and (3.1.5) can be rewritten as a pair of coupled wave equations:

$$\nabla^2 \vec{A} + k^2 \vec{A} = -\mu \vec{J} + \vec{\nabla} (\vec{\nabla} \cdot \vec{A} + i\omega \mu \epsilon \varphi), \quad (3.1.6)$$

$$\nabla^2 \varphi + k^2 \varphi = -\rho/\epsilon - i\omega (\vec{\nabla} \cdot \vec{A} + i\omega \mu \epsilon \varphi), \quad (3.1.7)$$

where $k^2 = \omega^2 \mu \epsilon$. The potentials are not unique: for any sufficiently smooth scalar function χ ,

$$\vec{A} \longrightarrow \vec{A} + \vec{\nabla} \chi, \quad \varphi \longrightarrow \varphi - i\omega \chi$$

leaves both $\vec{B} = \vec{\nabla} \times \vec{A}$ and $\vec{E} = -\vec{\nabla} \varphi - i\omega \vec{A}$ unchanged, since

$$\begin{aligned} \vec{\nabla} \times (\vec{A} + \vec{\nabla} \chi) &= \vec{\nabla} \times \vec{A}, \\ -\vec{\nabla}(\varphi - i\omega \chi) - i\omega(\vec{A} + \vec{\nabla} \chi) &= -\vec{\nabla} \varphi - i\omega \vec{A}. \end{aligned}$$

Vector and Scalar Potentials (cont.)

The gauge condition concerns both potentials. Define its residual by $Q = \vec{\nabla} \cdot \vec{A} + i\omega\mu\epsilon\varphi$. The transformation just verified gives

$$\begin{aligned} Q' &= \vec{\nabla} \cdot (\vec{A} + \vec{\nabla}\chi) + i\omega\mu\epsilon(\varphi - i\omega\chi) \\ &= Q + (\nabla^2 + k^2)\chi. \end{aligned}$$

Thus $Q' = 0$ is obtained by solving $(\nabla^2 + k^2)\chi = -Q$ with compatible regularity and boundary conditions. For localized Q in an unbounded homogeneous medium, the outgoing Green function constructed below gives $\chi(\vec{r}) = \int G(\vec{r}; \vec{r}') Q(\vec{r}') dv'$. A gauge transformation is admissible only when it respects the prescribed potential class; this is not an assertion of solvability for arbitrary additional boundary conditions on χ . In the radiation construction below, charge conservation directly verifies that the integral potentials already satisfy the Lorenz choice

$$\vec{\nabla} \cdot \vec{A} + i\omega\mu\epsilon\varphi = 0, \quad (3.1.8)$$

which is the Lorenz gauge condition. With this choice, the equations become decoupled:

$$\nabla^2 \vec{A} + k^2 \vec{A} = -\mu \vec{J}, \quad (3.1.9)$$

Vector and Scalar Potentials (cont.)

$$\nabla^2 \varphi + k^2 \varphi = -\frac{\rho}{\epsilon} = \frac{1}{i\omega\epsilon} \vec{\nabla} \cdot \vec{J}. \quad (3.1.10)$$

When electric charge and electric current are absent, $\vec{\nabla} \cdot \vec{D} = 0$ and, on the same topologically simple domain, it is similarly convenient to define an electric vector potential by

$$\vec{D} = -\vec{\nabla} \times \vec{F}.$$

The minus sign is the convention dual to $\vec{B} = +\vec{\nabla} \times \vec{A}$. Ampere's law then gives the intermediate steps

$$\vec{\nabla} \times \vec{H} = i\omega \vec{D} = -i\omega \vec{\nabla} \times \vec{F} \implies \vec{\nabla} \times (\vec{H} + i\omega \vec{F}) = 0,$$

and hence, in a simply connected region,

$$\vec{H} = -\vec{\nabla} \vartheta - i\omega \vec{F},$$

where ϑ is the magnetic scalar potential. The corresponding Lorenz condition is $\vec{\nabla} \cdot \vec{F} + i\omega\mu\epsilon \vartheta = 0$.

The Green Function Method

For the free-space radiation problem below, take $\omega > 0$, real positive μ and ϵ , and the outgoing branch $k = \omega\sqrt{\mu\epsilon} > 0$; then $\eta = \sqrt{\mu/\epsilon} > 0$. The same integral construction extends to a passive homogeneous medium after the corresponding non-growing outgoing branch is selected. Both the scalar potential equation (3.1.10) and the Cartesian components of the vector potential equation (3.1.9) are forms of the inhomogeneous scalar Helmholtz equation. A powerful method for solving such equations is the Green function technique. The Green function G is the fundamental solution, or impulse response, of the Helmholtz equation: the solution for a unit point source (a Dirac delta function):

$$\nabla^2 G + k^2 G = -\delta(\vec{r} - \vec{r}'). \quad (3.1.11)$$

Seek a rotationally invariant fundamental solution, with $R = |\vec{r} - \vec{r}'|$. Away from the source its equation is

$$\frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{dG}{dR} \right) + k^2 G = 0.$$

Writing $v = RG$ gives $v'' + k^2 v = 0$, and hence $G = (C_- e^{-ikR} + C_+ e^{ikR})/R$. With the convention $e^{i\omega t}$, the no-incoming-field boundary condition selects $C_+ = 0$. It remains to determine C_- . For a smooth compactly supported test function f , integrate Green's

The Green Function Method (cont.)

identity outside a sphere of radius δ about the source. The inner boundary has outward normal $-\hat{R}$, so

$$\begin{aligned}\langle (\nabla^2 + k^2)G, f \rangle &= \lim_{\delta \rightarrow 0} \int_{R=\delta} (f \partial_R G - G \partial_R f) dS \\ &= -4\pi C_- f(\vec{r}').\end{aligned}$$

Here $G = C_-/R + O(1)$, $\partial_R G = -C_-/R^2 + O(1)$, and the term containing $G \partial_R f$ vanishes in the limit. Comparison with (3.1.11) gives $C_- = 1/(4\pi)$:

$$G(\vec{r}; \vec{r}') = \frac{e^{-ikR}}{4\pi R}, \quad (3.1.12)$$

The normalization is therefore fixed by the delta source, whereas the exponential sign is selected by the radiation boundary condition. This form satisfies the *Sommerfeld radiation condition* at infinity, which requires that any radiating field behaves as an outgoing wave that decays with distance:

$$\lim_{R \rightarrow \infty} R(\partial_R G + ikG) = 0. \quad (3.1.13)$$

The Green Function Method (cont.)

The opposite exponential sign describes an incoming spherical wave and is excluded by the no-incoming-field boundary condition of the present radiation problem (although incoming waves are, of course, physical when an incident field is prescribed).

The delta function is defined such that:

$$\begin{aligned} \delta(\vec{r} - \vec{r}') &= 0, & \vec{r} \neq \vec{r}', \\ \int_V \delta(\vec{r} - \vec{r}') dv &= \begin{cases} 1, & \vec{r}' \text{ in the interior of } V, \\ 0, & \vec{r}' \text{ outside } V. \end{cases} \end{aligned} \quad (3.1.14)$$

From (3.1.14), equation (3.1.11) can be re-expressed as the following set of conditions on G :

$$\begin{aligned} \nabla^2 G + k^2 G &= 0, & \vec{r} \neq \vec{r}', \\ \int_V (\nabla^2 G + k^2 G) dv &= -1, & \vec{r}' \text{ in } V. \end{aligned} \quad (3.1.15)$$

We can also verify these conditions directly. Every volume integral below that contains the source is understood distributionally, as established by the test-function calculation above. In particular, the ordinary pointwise identity $\nabla^2 G = -k^2 G$ away from the source

The Green Function Method (cont.)

must not be integrated across $R = 0$ while omitting the delta contribution. First, for $\vec{r} \neq \vec{r}'$ (that is, $R \neq 0$),

$$\begin{aligned}\nabla^2 G &= \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial}{\partial R} \frac{e^{-ikR}}{4\pi R} \right) \\ &= -k^2 \frac{e^{-ikR}}{4\pi R} = -k^2 G.\end{aligned}\tag{3.1.16}$$

Thus, $\nabla^2 G + k^2 G = 0$ for $R \neq 0$. To verify the second condition, we integrate over an infinitesimal spherical volume V_0 centered at $\vec{r}'$ with radius R_0 :

$$\begin{aligned}\int_{V_0} \nabla^2 G dv &= \oint_{S_0} \vec{\nabla} G \cdot d\vec{s} = \oint_{S_0} (\partial_R G|_{R_0}) \hat{R} \cdot d\vec{s} \\ &= -e^{-ikR_0} (1 + ikR_0)\end{aligned}\tag{3.1.17}$$

The Green Function Method (cont.)

and

$$\begin{aligned}\int_{V_0} k^2 G dV &= \int_{V_0} \frac{k^2 e^{-ikR}}{4\pi R} R^2 \sin \theta d\theta d\phi dR \\&= k^2 \int_0^{R_0} R e^{-ikR} dR \\&= \frac{k^2}{-ik} \int_0^{R_0} R d(e^{-ikR}) \\&= ik \left[R_0 e^{-ikR_0} + \frac{1}{ik} (e^{-ikR_0} - 1) \right] \\&= e^{-ikR_0} (ikR_0 + 1) - 1.\end{aligned}\tag{3.1.18}$$

Adding these two results, we find that $\int_{V_0} (\nabla^2 G + k^2 G) dv = -1$, which confirms that our expression for the Green function is correct.

Uniqueness of the Outgoing Solution

Here $k > 0$ and the exterior medium is homogeneous with real positive ϵ and μ . We first establish the scalar statement used for potentials. Let u solve $(\nabla^2 + k^2)u = 0$ outside a containing sphere S_b , with $u = O(r^{-1})$ and $\partial_r u + iku = o(r^{-1})$ uniformly in angle.

Apply Green's second identity between S_b and a large sphere, using (3.1.12). The outer boundary term vanishes: writing it in terms of $\partial_r u + iku$ cancels the leading outgoing factors. With ∂_b directed out of the containing ball, the remaining representation is

$$u(\vec{r}) = \int_{S_b} [u(\vec{r}') \partial_b G(\vec{r}; \vec{r}') - G(\vec{r}; \vec{r}') \partial_b u(\vec{r}')] dS'.$$

In this formula $\vec{r}$ is outside S_b . The sign follows from the inner-boundary normal of the exterior domain, which is $-\hat{r}'$. On this fixed integration surface, $|\vec{r} - \vec{r}'| = r - \hat{r} \cdot \vec{r}' + O(r^{-1})$ and $1/|\vec{r} - \vec{r}'| = r^{-1} + O(r^{-2})$ uniformly. Expansion of the kernel and its derivatives therefore gives

$$u(\vec{r}) = \frac{e^{-ikr}}{r} u_\infty(\hat{r}) + O(r^{-2}),$$

Uniqueness of the Outgoing Solution (cont.)

also with the differentiated expansions needed below. This conclusion uses a finite-surface integral, not an exchange of a far-field limit with an unproved infinite modal expansion.

Suppose $u_\infty = 0$. Let Y_{nm} be the normalized scalar spherical harmonics whose completeness is proved in Appendix C.6.1, and set $u_{nm}(r) = \int u(r, \Omega) Y_{nm}^*(\Omega) d\Omega$. Projecting the scalar Helmholtz equation and integrating the angular Laplacian by parts gives

$$u_{nm}'' + \frac{2}{r} u_{nm}' + \left[k^2 - \frac{n(n+1)}{r^2} \right] u_{nm} = 0.$$

The outgoing condition excludes $h_n^{(1)}$ and hence $u_{nm} = c_{nm} h_n^{(2)}(kr)$. The exact finite formula derived in Appendix B.5.2 gives, at each fixed n ,

$$\lim_{r \rightarrow \infty} r e^{ikr} u_{nm}(r) = \frac{i^{n+1}}{k} c_{nm}.$$

On the other hand, the uniform far-field expansion makes this limit equal to $\int u_\infty Y_{nm}^* d\Omega = 0$. Thus every c_{nm} vanishes; scalar completeness gives $u = 0$ outside S_b . This argument takes the limit of each projection separately.

Uniqueness of the Outgoing Solution (cont.)

The vanishing extends through a connected homogeneous source-free region. To see why, enclose any interior point by a small source-free ball and use its interior Green representation. Its kernels are real analytic in the observation point on every smaller ball, since the observation point stays separated from the integration surface. The integral consequently has a convergent local power series. A solution that vanishes on an open part of one such ball vanishes on the ball by analyticity. Overlapping balls propagate that statement throughout the connected region.

For a scalar solution regular in all space, Green's first identity gives

$$\int_{S_R} u^* \partial_r u \, dS = \int_{B_R} (|\vec{\nabla} u|^2 - k^2 |u|^2) \, dv,$$

whose right-hand side is real. Its imaginary part at infinity is $-k \int |u_\infty|^2 d\Omega$. Hence $u_\infty = 0$, and the preceding argument proves that a regular all-space outgoing homogeneous scalar solution is zero. Applied componentwise, this proves the corresponding uniqueness statement for the vector potential.

For electromagnetic exterior uniqueness, consider the difference of two solutions outside a bounded obstacle, with homogeneous PEC or PMC data (or complementary homogeneous tangential data on a regular boundary partition). Assume a connected

Uniqueness of the Outgoing Solution (cont.)

homogeneous exterior and the outgoing conditions above for the Cartesian field components. Maxwell's equations imply a scalar Helmholtz equation for each component. Their leading far-field terms satisfy

$$\hat{r} \cdot \vec{E}_\infty = 0, \quad \vec{H}_\infty = \frac{1}{\eta} \hat{r} \times \vec{E}_\infty.$$

These follow respectively from $\vec{\nabla} \cdot \vec{E} = 0$ and from $\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H}$, using $\vec{\nabla}(e^{-ikr}/r) = -ik\hat{r}e^{-ikr}/r + O(r^{-2})$. The real part of Poynting's identity is zero in the lossless exterior. The homogeneous boundary data make the obstacle-boundary flux zero, so

$$0 = \lim_{R \rightarrow \infty} \frac{1}{2} \operatorname{Re} \int_{S_R} (\vec{E} \times \vec{H}^*) \cdot \hat{r} dS = \frac{1}{2\eta} \int |\vec{E}_\infty|^2 d\Omega.$$

Thus $\vec{E}_\infty = 0$. The scalar result makes every Cartesian component of $\vec{E}$ vanish, and the curl equation then gives $\vec{H} = 0$. This proves uniqueness for the stated exterior class; it does not assert the same theorem for arbitrary inhomogeneous, active, waveguide, or nonstandard boundary-value problems.

The General Radiation Solution

The solution of the wave equation (3.1.9) follows by superposition. Using the sifting property (3.1.14), the source itself is a continuous superposition of point sources,

$$\mu \vec{J}(\vec{r}) = \int_{V'} \mu \vec{J}(\vec{r}') \delta(\vec{r} - \vec{r}') dv'.$$

The infinitesimal point source at $\vec{r}'$ has vector strength

$$d\vec{S}(\vec{r}') = \mu \vec{J}(\vec{r}') dv'.$$

By (3.1.11), its contribution to the response is

$$d\vec{A}(\vec{r}) = \mu \vec{J}(\vec{r}') G(\vec{r}; \vec{r}') dv'.$$

Since $\nabla^2 + k^2$ is linear and acts on the observation coordinate, it passes under the source integral, so the response to the distribution is the sum of these contributions.

The General Radiation Solution (cont.)

Equivalently, multiplying (3.1.11) by $\mu \vec{J}(\vec{r}')$ and integrating over a volume V' containing all sources gives

$$\begin{aligned} (\nabla^2 + k^2) \int_{V'} \mu \vec{J}(\vec{r}') G(\vec{r}; \vec{r}') dv' \\ = \int_{V'} \mu \vec{J}(\vec{r}') [(\nabla^2 + k^2) G(\vec{r}; \vec{r}')] dv' \\ = - \int_{V'} \mu \vec{J}(\vec{r}') \delta(\vec{r} - \vec{r}') dv' = -\mu \vec{J}(\vec{r}), \end{aligned}$$

which is exactly the wave equation (3.1.9); the solution for $\vec{A}(\vec{r})$ is thus the convolution of the source distribution with the Green function. Here the intended problem is radiation by localized sources in an otherwise homogeneous, unbounded medium, with no prescribed incident field. Any second solution differs from this integral by a field $\vec{A}_0$ satisfying $\nabla^2 \vec{A}_0 + k^2 \vec{A}_0 = 0$ everywhere, regular at finite points, and with the homogeneous outgoing condition. The all-space scalar argument in Section 3.1.3, applied to each Cartesian component, gives $\vec{A}_0 = 0$. This is a statement about potentials satisfying the same Lorenz-gauge wave equation and the same regularity and

The General Radiation Solution (cont.)

radiation conditions; it is not a denial of gauge freedom in a larger potential class. Without the regularity and no-incident-field assumptions, a homogeneous background potential could be added. Under the stated radiation assumptions, the unique solution is

$$\vec{A}(\vec{r}) = \int_{V'} \mu \vec{J}(\vec{r}') G(\vec{r}; \vec{r}') dv'. \quad (3.1.19)$$

Similarly, the continuity equation for the $e^{+i\omega t}$ convention is

$$\vec{\nabla}' \cdot \vec{J}(\vec{r}') = -i\omega\rho(\vec{r}'), \quad \rho = -\frac{1}{i\omega} \vec{\nabla}' \cdot \vec{J}.$$

Here $\vec{\nabla}$ acts on the observation coordinate $\vec{r}$, whereas $\vec{\nabla}'$ acts on the source coordinate $\vec{r}'$. Therefore (3.1.11) and (3.1.10) give the equivalent scalar-potential forms

$$\varphi(\vec{r}) = \frac{1}{\epsilon} \int_{V'} \rho(\vec{r}') G dv' = -\frac{1}{i\omega\epsilon} \int_{V'} (\vec{\nabla}' \cdot \vec{J}(\vec{r}')) G dv'. \quad (3.1.20)$$

The General Radiation Solution (cont.)

For localized sources, integrate over all source space, or take V' with $\vec{J} = 0$ on its boundary. Since $\vec{\nabla} G = -\vec{\nabla}' G$, integration by parts gives

$$\begin{aligned}\vec{\nabla} \cdot \vec{A} &= -\mu \int_{V'} \vec{J} \cdot \vec{\nabla}' G dv' \\ &= \mu \int_{V'} (\vec{\nabla}' \cdot \vec{J}) G dv' \\ &= -i\omega\mu \int_{V'} \rho G dv' = -i\omega\mu\epsilon\varphi.\end{aligned}$$

This establishes the gauge condition for the constructed potentials. For distributional surface or line sources the same identity is read in the weak sense, with the complete conserved source included. Differentiation with respect to $\vec{r}$ may be moved under a nonsingular source integral, or interpreted distributionally at a source. The two terms of (3.1.3) are then

$$\begin{aligned}-\vec{\nabla}\varphi &= \frac{1}{i\omega\epsilon} \int_{V'} (\vec{\nabla}' \cdot \vec{J}) \vec{\nabla} G dv', \\ -i\omega\vec{A} &= -i\omega\mu \int_{V'} \vec{J} G dv' .\end{aligned}$$

The General Radiation Solution (cont.)

Using

$$\omega\mu = k\eta, \quad \frac{1}{i\omega\epsilon} = -\frac{i}{\omega\epsilon} = -\frac{ik\eta}{k^2},$$

and adding the two contributions gives the electric radiation integral

$$\begin{aligned} \vec{E}(\vec{r}) = -ik\eta \int_{V'} \left[\vec{J}(\vec{r}') G(\vec{r}; \vec{r}') \right. \\ \left. + \frac{1}{k^2} (\vec{\nabla}' \cdot \vec{J}(\vec{r}')) \vec{\nabla} G(\vec{r}; \vec{r}') \right] dv'. \end{aligned} \quad (3.1.21)$$

For the magnetic field, the product rule and $\vec{\nabla} \times (\vec{J}G) = \vec{\nabla} G \times \vec{J} = -\vec{J} \times \vec{\nabla} G$ give every step:

$$\begin{aligned} \vec{H}(\vec{r}) &= \frac{1}{\mu} \vec{\nabla} \times \vec{A} = \int_{V'} \vec{\nabla} \times (\vec{J}(\vec{r}') G) dv' \\ &= - \int_{V'} \vec{J}(\vec{r}') \times \vec{\nabla} G(\vec{r}; \vec{r}') dv'. \end{aligned} \quad (3.1.22)$$

The General Radiation Solution (cont.)

It is convenient to define the following integral operators:

$$\mathfrak{L}\{\vec{X}\} = -ik \int_{V'} \left[\vec{X}G + \frac{1}{k^2}(\vec{\nabla}' \cdot \vec{X})\vec{\nabla}G \right] dv', \quad (3.1.23)$$

$$\mathfrak{K}\{\vec{X}\} = - \int_{V'} \vec{X} \times \vec{\nabla}G dv'. \quad (3.1.24)$$

This expanded form keeps the operator placement visible: $\vec{\nabla}'$ differentiates the source distribution, whereas $\vec{\nabla}$ differentiates G . A notation such as $\vec{\nabla}[(\vec{\nabla}' \cdot \vec{X})G]$ is equivalent only when it is understood that the observation gradient passes through the source-only factor and acts on G :

$$\vec{\nabla} [(\vec{\nabla}' \cdot \vec{X})G] = (\vec{\nabla}' \cdot \vec{X})\vec{\nabla}G.$$

To make the duality signs explicit, write Maxwell's curl equations with both source types as

$$\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H} - \vec{M}, \quad \vec{\nabla} \times \vec{H} = i\omega\epsilon\vec{E} + \vec{J}.$$

Let $(\vec{E}_X, \vec{H}_X)$ be the fields generated by an electric current $\vec{X}$ with no magnetic current. Define

$$\vec{E}_d = -\eta\vec{H}_X, \quad \vec{H}_d = \frac{1}{\eta}\vec{E}_X, \quad \vec{M}_d = \eta\vec{X}, \quad \vec{J}_d = \vec{0}.$$

The General Radiation Solution (cont.)

Using $\eta\epsilon = \mu/\eta = \sqrt{\mu\epsilon}$, direct substitution gives

$$\vec{\nabla} \times \vec{E}_d = -\eta \left(i\omega\epsilon \vec{E}_X + \vec{X} \right) = -i\omega\mu \vec{H}_d - \vec{M}_d,$$

$$\vec{\nabla} \times \vec{H}_d = \frac{1}{\eta} \left(-i\omega\mu \vec{H}_X \right) = i\omega\epsilon \vec{E}_d.$$

For a prescribed magnetic current $\vec{M}$, choose $\vec{X} = \vec{M}/\eta$. Inserting the electric-source fields $\vec{E}_X = \eta \mathcal{L}\{\vec{X}\}$ and $\vec{H}_X = \mathcal{R}\{\vec{X}\}$ then yields the magnetic-source formulas below without an implicit sign choice. Using these operators, the electric and magnetic fields generated by electric and magnetic sources can be expressed in a compact and symmetric form. For electric sources:

$$\vec{E} = \eta \mathcal{L}\{\vec{J}\}, \quad (3.1.25)$$

$$\vec{H} = \mathcal{R}\{\vec{J}\}, \quad (3.1.26)$$

and by applying the duality transform, for magnetic sources:

$$\vec{E} = -\mathcal{R}\{\vec{M}\}, \quad (3.1.27)$$

The General Radiation Solution (cont.)

$$\vec{H} = \frac{1}{\eta} \mathcal{L}\{\vec{M}\}. \quad (3.1.28)$$

By the principle of superposition, the total fields from both electric and magnetic sources are:

$$\vec{E} = \eta \mathcal{L}\{\vec{J}\} - \mathcal{R}\{\vec{M}\}, \quad (3.1.29)$$

$$\vec{H} = \mathcal{R}\{\vec{J}\} + \frac{1}{\eta} \mathcal{L}\{\vec{M}\}. \quad (3.1.30)$$

An alternative expression follows directly from the Lorenz condition:

$$\varphi = -\frac{1}{i\omega\mu\epsilon} \vec{\nabla} \cdot \vec{A},$$

and therefore

$$\begin{aligned} \vec{E} &= -i\omega\vec{A} + \frac{1}{i\omega\mu\epsilon} \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) \\ &= -i\omega \left(1 + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \cdot \right) \vec{A}. \end{aligned}$$

The General Radiation Solution (cont.)

Substitution of (3.1.19) and use of $\omega\mu = k\eta$ now gives

$$\vec{E}(\vec{r}) = -ik\eta \int_{V'} \left(1 + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \cdot\right) [\vec{J}(\vec{r}') G(\vec{r}; \vec{r}')] dv'. \quad (3.1.31)$$

For observation points strictly outside the source support, (3.1.31) is an ordinary integral. Within a volume-current source, it must be interpreted distributionally, or as a specified principal-volume integral together with its local source term. It is important to understand the distinction between the two expressions for the electric field, (3.1.21) and (3.1.31). In (3.1.21), $\vec{\nabla}'$ differentiates the source current and $\vec{\nabla}$ differentiates only the Green function. This results in a weaker singularity in the integrand. In contrast, in (3.1.31), both $\vec{\nabla}$ operators act on the field point $\vec{r}$, and therefore on the Green function, leading to a higher-order singularity. This form is typically more suitable for far-field calculations, where simplifying approximations can be made.

The Stratton-Chu Formulation

Based on the surface equivalence principle (Section 1.5.4), let a closed surface S_0 lie in a homogeneous, source-free neighborhood and enclose all actual sources. Choose $\hat{n}$ to point from the extinguished interior into the exterior region of interest. Setting the equivalent interior fields to zero replaces the original sources by

$$\begin{cases} \vec{J}_s = \hat{n} \times \vec{H}, \\ \vec{M}_s = -\hat{n} \times \vec{E}, \end{cases} \quad (3.1.32)$$

where the fields on the right are the original exterior limiting fields. For an observation point strictly outside S_0 , the general radiation solution (3.1.29) gives

$$\begin{aligned} \vec{E} &= \eta \mathcal{L}\{\vec{J}_s\} - \mathcal{R}\{\vec{M}_s\} \\ &= -ik\eta \oint_{S_0} \left[\vec{J}_s G + \frac{1}{k^2} (\vec{\nabla}'_s \cdot \vec{J}_s) \vec{\nabla} G \right] ds' \\ &\quad + \oint_{S_0} \vec{M}_s \times \vec{\nabla} G ds'. \end{aligned} \quad (3.1.33)$$

The Stratton-Chu Formulation (cont.)

Here $\vec{\nabla}'_s \cdot$ is the surface divergence. Because there is no normal volume-current flux on either side of the equivalent sheet, surface charge conservation and the normal- $\vec{D}$ matching condition (1.2.8) give, step by step,

$$\vec{\nabla}'_s \cdot \vec{J}_s + i\omega\rho_s = 0, \quad \rho_s = \hat{n} \cdot (\vec{D}_{\text{out}} - \vec{D}_{\text{in}}) = \epsilon \hat{n} \cdot \vec{E}, \quad (3.1.34)$$

or

$$\vec{\nabla}'_s \cdot \vec{J}_s = -i\omega\epsilon(\hat{n} \cdot \vec{E}).$$

Substituting (3.1.32) and (3.1.34) into (3.1.33) and using the property $\vec{\nabla}' G = -\vec{\nabla} G$, we arrive at the expression for the electric field:

$$\begin{aligned} -\frac{ik\eta}{k^2} [-i\omega\epsilon(\hat{n} \cdot \vec{E})] \vec{\nabla} G &= -(\hat{n} \cdot \vec{E}) \vec{\nabla} G = (\hat{n} \cdot \vec{E}) \vec{\nabla}' G, \\ \vec{M}_s \times \vec{\nabla} G &= (-\hat{n} \times \vec{E}) \times (-\vec{\nabla}' G) \\ &= (\hat{n} \times \vec{E}) \times \vec{\nabla}' G, \end{aligned}$$

The Stratton-Chu Formulation (cont.)

where $k\eta = \omega\mu$ and $k^2 = \omega^2\mu\epsilon$ make the scalar coefficient in the first line equal to -1 . Therefore

$$\begin{aligned}\vec{E} = \oint_{S_0} & \left[-ik\eta (\hat{n} \times \vec{H}) G + (\hat{n} \cdot \vec{E}) \vec{\nabla}' G \right. \\ & \left. + (\hat{n} \times \vec{E}) \times \vec{\nabla}' G \right] ds'.\end{aligned}\tag{3.1.35}$$

By applying the duality transform, we can obtain the corresponding expression for the magnetic field:

$$\begin{aligned}\vec{H} = \oint_{S_0} & \left[i\frac{k}{\eta} (\hat{n} \times \vec{E}) G + (\hat{n} \cdot \vec{H}) \vec{\nabla}' G \right. \\ & \left. + (\hat{n} \times \vec{H}) \times \vec{\nabla}' G \right] ds'.\end{aligned}\tag{3.1.36}$$

This pair of equations constitutes the *Stratton–Chu formulation*, which provides a rigorous and general framework for solving radiation problems. Notably, the Stratton–Chu formulation explicitly involves the normal components of the electric and magnetic fields.

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The Far-Field Approximation

In the far-field region, the observation distance r is much larger than the source size $a = \max |\vec{r}'|$, and the observation point is many wavelengths away, $kr \gg 1$. For an extended source, the phase approximation below also requires $ka^2/r \ll 1$, the Fraunhofer condition. The condition $r \gg a$ controls the amplitude expansion, while $kr \gg 1$ controls the neglect of higher-order $1/r$ field terms. The denominator of the Green function (3.1.12) can be approximated as $4\pi r$, and the phase term in the numerator can be expanded as

$$\begin{aligned} R = |\vec{r} - \vec{r}'| &= \sqrt{r^2 - 2r(\hat{r} \cdot \vec{r}') + r'^2} \\ &= r \left(1 - 2\frac{\hat{r} \cdot \vec{r}'}{r} + \frac{r'^2}{r^2} \right)^{1/2}. \end{aligned}$$

Using the generalized binomial theorem with small argument x ,

$$(1 + x)^{1/2} \approx 1 + \frac{1}{2}x,$$

and retaining only the first-order term, we obtain

$$R \approx r - \hat{r} \cdot \vec{r}'. \quad (3.2.1)$$

The Far-Field Approximation (cont.)

The approximation (3.2.1) is first order in the source-size ratio a/r . If the next term is needed, the binomial expansion must be carried one order further. With

$$x = -2\frac{\hat{r} \cdot \vec{r}'}{r} + \frac{r'^2}{r^2},$$

substitution into $(1+x)^{1/2} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2$ gives, after multiplying by r ,

$$R \approx r - \hat{r} \cdot \vec{r}' + \frac{r'^2 - (\hat{r} \cdot \vec{r}')^2}{2r} + \dots,$$

so the term dropped in (3.2.1) is $O(a^2/r)$. The amplitude factor expands as

$$\frac{1}{R} = \frac{1}{r} \left[1 + \frac{\hat{r} \cdot \vec{r}'}{r} + O\left(\frac{a^2}{r^2}\right) \right].$$

Its relative correction is $O(a/r)$, so $1/R$ may be replaced by $1/r$ when $r \gg a$. The phase term e^{-ikR} is more delicate, because the error in R is multiplied by k . The omitted

The Far-Field Approximation (cont.)

quadratic term contributes a phase error of order ka^2/r , so the linear phase approximation is valid when $ka^2/r \ll 1$. Under this condition,

$$e^{-ikR} \approx e^{-ikr(1-\hat{r}\cdot\vec{r}'/r)} \approx e^{-ikr} e^{ik\hat{r}\cdot\vec{r}'} \quad (3.2.2)$$

This approximation preserves the leading angular dependence of the radiation pattern. Thus, in the far field, the Green function simplifies to

$$G(\vec{r}; \vec{r}') \approx \frac{e^{-ikr}}{4\pi r} e^{ik\hat{r}\cdot\vec{r}'} = G_r(r) G_a(\theta, \phi), \quad (3.2.3)$$

where $G_r(r) = e^{-ikr}/(4\pi r)$ represents the radial dependence, and $G_a(\theta, \phi) = e^{ik\hat{r}\cdot\vec{r}'}$ represents the angular dependence.

Here $k = 2\pi/\lambda$, with λ the wavelength. To apply (3.2.3) to the radiation integral, differentiate both factors without suppressing their orders. With $\bar{\bar{I}}$ the identity dyad, $\vec{\nabla}\vec{r} = \vec{\nabla}(r\hat{r}) = \vec{\nabla}(r)\hat{r} + r\vec{\nabla}\hat{r} = \hat{r}\hat{r} + r\vec{\nabla}\hat{r} = \bar{\bar{I}}$, so

$$\vec{\nabla}\hat{r} = \frac{\bar{\bar{I}} - \hat{r}\hat{r}}{r}, \quad \vec{\nabla}(\hat{r} \cdot \vec{r}') = \frac{\bar{\bar{I}} - \hat{r}\hat{r}}{r} \cdot \vec{r}',$$

The Far-Field Approximation (cont.)

we have

$$\begin{aligned}\vec{\nabla} G_r &= \left(-ik - \frac{1}{r}\right) G_r \hat{r}, \\ \vec{\nabla} G_a &= \frac{ik}{r} G_a \left(\vec{l} - \hat{r}\hat{r}\right) \cdot \vec{r}'.\end{aligned}$$

The second line has magnitude at most $ka|G_a|/r$. The product rule gives an exact identity for the approximate kernel $G_F = G_r G_a$:

$$\vec{\nabla} G_F = G_F \left[-ik\hat{r} - \frac{\hat{r}}{r} + \frac{ik}{r}(\vec{l} - \hat{r}\hat{r}) \cdot \vec{r}'\right].$$

To control the error for the exact kernel $G = e^{-ikR}/(4\pi R)$, put $\hat{R} = (\vec{r} - \vec{r}')/R$. Since $\vec{\nabla} R = \hat{R}$ and $\vec{\nabla} \hat{R} = (\vec{l} - \hat{R}\hat{R})/R$, differentiation gives

$$\begin{aligned}\vec{\nabla} G &= \left(-ik - \frac{1}{R}\right) G \hat{R}, \\ \vec{\nabla} \vec{\nabla} G &= G \left[-k^2 \hat{R}\hat{R} + \left(\frac{ik}{R} + \frac{1}{R^2}\right) (3\hat{R}\hat{R} - \vec{l})\right].\end{aligned}$$

The Far-Field Approximation (cont.)

For $|\vec{r}'| \leq a$ and $r > 2a$, expanding the displacement gives

$$\hat{R} = \hat{r} - \frac{(\bar{\vec{l}} - \hat{r}\hat{r}) \cdot \vec{r}'}{r} + O\left(\frac{a^2}{r^2}\right), \quad \frac{1}{R} = \frac{1}{r} + O\left(\frac{a}{r^2}\right).$$

Thus the exact kernel satisfies

$$\begin{aligned} \vec{\nabla} G = G \bigg[& -ik\hat{r} - \frac{\hat{r}}{r} + \frac{ik}{r}(\bar{\vec{l}} - \hat{r}\hat{r}) \cdot \vec{r}' \\ & + O\left(\frac{a + ka^2}{r^2}\right) \bigg]. \end{aligned} \tag{3.2.4}$$

Define the exact remainder by

$$\vec{\nabla} G = -ikG\hat{r} + \vec{\Delta}, \quad |\vec{\Delta}| = |G| O\left(\frac{1 + ka}{r}\right).$$

The Far-Field Approximation (cont.)

Because $G = O(1/r)$, the absolute error is $O((1 + ka)/r^2)$; relative to $k|G|$, it vanishes when $kr \gg 1$ and $r \gg a$. Using the exact Hessian and $\vec{\nabla} \vec{\Delta} = \vec{\nabla} \vec{\nabla} G + ik[(\vec{\nabla} G)\hat{r} + G\vec{\nabla} \hat{r}]$ also gives

$$|\vec{\nabla} \vec{\Delta}| = |G| O\left(\frac{k(1 + ka)}{r} + \frac{(1 + ka)^2}{r^2}\right).$$

Both estimates are uniform over the bounded source and justify the second differentiation below. Because $\vec{J}(\vec{r}')$ is independent of the observation coordinate,

$$\begin{aligned}\vec{\nabla} \vec{\nabla} \cdot [\vec{J}G] &= \vec{\nabla}[\vec{J} \cdot \vec{\nabla} G] \\ &= \vec{\nabla}[-ik(\hat{r} \cdot \vec{J})G + \vec{J} \cdot \vec{\Delta}] \\ &= -ik\{(\hat{r} \cdot \vec{J})\vec{\nabla} G + G\vec{\nabla}(\hat{r} \cdot \vec{J})\} + \vec{\nabla}(\vec{J} \cdot \vec{\Delta}).\end{aligned}\tag{3.2.5}$$

To proceed with this derivation, we first note the following vector identity involving the unit dyadic $\vec{\vec{I}}$:

$$\begin{aligned}\vec{\nabla} \vec{r} &= (\hat{x}\partial_x + \hat{y}\partial_y + \hat{z}\partial_z)(\hat{x}\hat{x} + \hat{y}\hat{y} + \hat{z}\hat{z}) \\ &= \hat{x}\hat{x} + \hat{y}\hat{y} + \hat{z}\hat{z} = \vec{\vec{I}},\end{aligned}\tag{3.2.6}$$

The Far-Field Approximation (cont.)

where a dyadic is a tensor of rank two, which can be represented as a square matrix:

$$\begin{aligned}\bar{\bar{F}} &= F_{xx}\hat{x}\hat{x} + F_{yx}\hat{y}\hat{x} + F_{zx}\hat{z}\hat{x} \\ &\quad + F_{xy}\hat{x}\hat{y} + F_{yy}\hat{y}\hat{y} + F_{zy}\hat{z}\hat{y} \\ &\quad + F_{xz}\hat{x}\hat{z} + F_{yz}\hat{y}\hat{z} + F_{zz}\hat{z}\hat{z} \\ &= \begin{bmatrix} F_{xx} & F_{xy} & F_{xz} \\ F_{yx} & F_{yy} & F_{yz} \\ F_{zx} & F_{zy} & F_{zz} \end{bmatrix}. \end{aligned} \tag{3.2.7}$$

The juxtaposition of two vectors, such as $\bar{\bar{F}} = \vec{a}\vec{b}$, forms a dyadic product with components $F_{mn} = a_m b_n$. We have the following rules for dyadic calculations:

- $\vec{c} \cdot (\vec{a}\vec{b}) = (\vec{c} \cdot \vec{a})\vec{b}$
- $(\vec{a}\vec{b}) \cdot \vec{c} = \vec{a}(\vec{b} \cdot \vec{c})$
- $\vec{c} \times (\vec{a}\vec{b}) = (\vec{c} \times \vec{a})\vec{b}$
- $(\vec{a}\vec{b}) \times \vec{c} = \vec{a}(\vec{b} \times \vec{c})$

The Far-Field Approximation (cont.)

In particular, $\vec{\nabla}\hat{r} = (\vec{\bar{I}} - \hat{r}\hat{r})/r$, as derived above. The term $\vec{\nabla}[\hat{r} \cdot \vec{J}(\vec{r}')]]$ in (3.2.5) can be simplified using the vector identity

$$\vec{\nabla}(\vec{a} \cdot \vec{b}) = \vec{a} \times (\vec{\nabla} \times \vec{b}) + \vec{b} \times (\vec{\nabla} \times \vec{a}) + (\vec{a} \cdot \vec{\nabla})\vec{b} + (\vec{b} \cdot \vec{\nabla})\vec{a}:$$

$$\begin{aligned}\vec{\nabla} [\hat{r} \cdot \vec{J}(\vec{r}')] &= [\vec{J}(\vec{r}') \cdot \vec{\nabla}] \hat{r} = \vec{J}(\vec{r}') \cdot \vec{\nabla} \hat{r} \\ &= \frac{\vec{J} - J_r \hat{r}}{r} = \frac{J_\theta \hat{\theta} + J_\phi \hat{\phi}}{r} = O\left(\frac{1}{r}\right).\end{aligned}\tag{3.2.8}$$

Thus, writing $J_r = \hat{r} \cdot \vec{J}$ for the radial component and $\vec{J}_T = \vec{J} - J_r \hat{r}$ for the transverse part, the expression in (3.2.5) becomes

$$\begin{aligned}\vec{\nabla} \vec{\nabla} \cdot [\vec{J}G] &= -ik \left\{ J_r (-ikG\hat{r} + \vec{\Delta}) + G \frac{\vec{J}_T}{r} \right\} + \vec{\nabla}(\vec{J} \cdot \vec{\Delta}) \\ &= -k^2 J_r \hat{r} G \\ &\quad + O\left(\frac{kG(1+ka)|\vec{J}|}{r}\right) + O\left(\frac{G(1+ka)^2|\vec{J}|}{r^2}\right).\end{aligned}\tag{3.2.9}$$

The Far-Field Approximation (cont.)

Relative to the retained $k^2 G[\vec{J}]$ term, the displayed errors are of order $(1 + ka)/(kr)$ and its square; they vanish under the far-field conditions stated above. Applying these approximations, the expression for the electric field in (3.1.31) simplifies to:

$$\begin{aligned}\vec{E} &\approx -ik\eta \int_{V'} G [\vec{J}(\vec{r}') - J_r \hat{r}] dv' \\ &= -ik\eta \frac{e^{-ikr}}{4\pi r} \int_{V'} (J_\theta \hat{\theta} + J_\phi \hat{\phi}) e^{ik\hat{r} \cdot \vec{r}'} dv' \\ &= -ik\eta \frac{e^{-ikr}}{4\pi r} \int_{V'} (\hat{\theta} \hat{\theta} + \hat{\phi} \hat{\phi}) \cdot \vec{J}(\vec{r}') e^{ik\hat{r} \cdot \vec{r}'} dv' \\ &= ik\eta \frac{e^{-ikr}}{4\pi r} \int_{V'} \hat{r} \times [\hat{r} \times \vec{J}(\vec{r}')] e^{ik\hat{r} \cdot \vec{r}'} dv'.\end{aligned}\tag{3.2.10}$$

From (3.1.22), the magnetic field in the far-field region is given by:

$$\vec{H} \approx -ik \frac{e^{-ikr}}{4\pi r} \int_{V'} [\hat{r} \times \vec{J}(\vec{r}')] e^{ik\hat{r} \cdot \vec{r}'} dv' = \frac{1}{\eta} \hat{r} \times \vec{E}.\tag{3.2.11}$$

The Far-Field Approximation (cont.)

For the general case including both electric and magnetic sources, we can apply the duality principle to obtain the far-field expression for the electric field and the magnetic field:

$$\vec{E} \approx -ik \frac{e^{-ikr}}{4\pi r} \int_{V'} [\eta (\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{J}(\vec{r}') - \hat{r} \times \vec{M}(\vec{r}')] e^{ik\hat{r} \cdot \vec{r}'} dv' \quad (3.2.12)$$

$$\vec{H} \approx -ik \frac{e^{-ikr}}{4\pi r} \int_{V'} \left[\frac{1}{\eta} (\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{M}(\vec{r}') + \hat{r} \times \vec{J}(\vec{r}') \right] e^{ik\hat{r} \cdot \vec{r}'} dv'. \quad (3.2.13)$$

If the spatial Fourier transform is defined by

$$\tilde{J}(\vec{q}) = \int \vec{J}(\vec{r}') e^{i\vec{q} \cdot \vec{r}'} dv',$$

then the electric-current far field samples its transverse projection on the sphere $\vec{q} = k\hat{r}$. With the opposite sign convention this same integral is called the inverse transform; the physically invariant statement is the sampling and transverse projection, not the

The Far-Field Approximation (cont.)

transform name. The three equalities in the chain leading to (3.2.12) rest on the resolution of the source current into its radial and transverse parts. Writing $\vec{J}(\vec{r}') = J_r \hat{r} + J_\theta \hat{\theta} + J_\phi \hat{\phi}$ with $J_r = \hat{r} \cdot \vec{J}$, the radial component is removed by subtraction,

$$\vec{J}(\vec{r}') - J_r \hat{r} = J_\theta \hat{\theta} + J_\phi \hat{\phi},$$

which is the purely transverse current already encountered in the simplification of the radiation integral. Recalling that the unit dyadic resolves in the spherical basis as

$\bar{\bar{1}} = \hat{r}\hat{r} + \hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}$, this transverse current is the action of the projection dyadic $\bar{\bar{1}} - \hat{r}\hat{r} = \hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}$ on $\vec{J}$; using the dyadic rule $(\vec{a}\vec{b}) \cdot \vec{c} = \vec{a}(\vec{b} \cdot \vec{c})$,

$$\begin{aligned}(\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{J} &= \hat{\theta}(\hat{\theta} \cdot \vec{J}) + \hat{\phi}(\hat{\phi} \cdot \vec{J}) \\ &= J_\theta \hat{\theta} + J_\phi \hat{\phi}.\end{aligned}$$

Finally, the same transverse current may be written as a double cross product. Applying the identity $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$ with $\vec{a} = \vec{b} = \hat{r}$ and $\hat{r} \cdot \hat{r} = 1$,

$$\begin{aligned}\hat{r} \times (\hat{r} \times \vec{J}) &= \hat{r}(\hat{r} \cdot \vec{J}) - \vec{J} = J_r \hat{r} - \vec{J} \\ &= -(\hat{\theta}\hat{\theta} + \hat{\phi}\hat{\phi}) \cdot \vec{J},\end{aligned}$$

The Far-Field Approximation (cont.)

so the transverse projection equals $-\hat{r} \times (\hat{r} \times \vec{J})$, and the sign of the prefactor reverses from $-ik\eta$ to $+ik\eta$ in the last line.

The mixed-source form (3.2.12) then follows by duality. For magnetic sources $\vec{E} = -\Re\{\vec{M}\}$, and with $\vec{\nabla} G \approx -ikG\hat{r}$ in the far field,

$$\begin{aligned}\Re\{\vec{M}\} &= - \int_{V'} \vec{M} \times \vec{\nabla} G \, dv' \approx ik \int_{V'} (\vec{M} \times \hat{r}) G \, dv' \\ &= -ik \frac{e^{-ikr}}{4\pi r} \int_{V'} (\hat{r} \times \vec{M}) e^{ik\hat{r} \cdot \vec{r}'} \, dv' .\end{aligned}$$

Hence $\vec{E} = -\Re\{\vec{M}\}$ contributes a term $+ik(e^{-ikr}/4\pi r)\hat{r} \times \vec{M}$, which appears as $-\hat{r} \times \vec{M}$ inside the bracket of (3.2.12) once the common factor $-ik$ is pulled out; superposing this with the electric contribution, and repeating the argument with the roles of $\vec{J}$ and $\vec{M}$ exchanged, yields (3.2.13).

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Radiation from a Hertzian Dipole

The Hertzian dipole is the ideal point-current model of an electrically short, uniform-current element. Let its effective length be denoted by dl , let q be the endpoint-charge phasor, and let positive current flow toward the endpoint carrying $+q$. The time-domain relation $I(t) = dq(t)/dt$ becomes, under the $e^{+i\omega t}$ convention,

$$I = i\omega q, \quad \vec{p} = q dl \hat{z}, \quad I dl \hat{z} = i\omega \vec{p}.$$

The point-source distribution at the origin is

$$\vec{J}(\vec{r}') = I dl \hat{z} \delta(\vec{r}'), \quad \int \vec{J}(\vec{r}') dv' = I dl \hat{z}.$$

Equivalently, before the infinitesimal limit a filament on the z -axis has $\vec{J} = I \hat{z} \delta(x') \delta(y')$ over the element, so $\vec{J} dv' = I \hat{z} dz'$; integration over its length produces $I dl \hat{z}$. The associated endpoint charges are required by the continuity equation, so the ideal current element is not a current source in isolation. Substitution into (3.1.31) gives

$$\vec{E}(\vec{r}) = -ik\eta dl \left(1 + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \cdot \right) \hat{z} G. \quad (3.3.1)$$

Radiation from a Hertzian Dipole (cont.)

and (3.1.22) gives

$$\vec{H}(\vec{r}) = -Idl \hat{z} \times \vec{\nabla} G. \quad (3.3.2)$$

To evaluate the derivatives in spherical coordinates, first differentiate $G = e^{-ikr}/(4\pi r)$ explicitly:

$$\begin{aligned} \partial_r G &= \frac{1}{4\pi} \left[(-ik)e^{-ikr} \frac{1}{r} + e^{-ikr} \left(-\frac{1}{r^2} \right) \right] \\ &= -\left(ik + \frac{1}{r} \right) G, \end{aligned} \quad (3.3.3)$$

$$\vec{\nabla} G = \hat{r} \partial_r G = -\left(ik + \frac{1}{r} \right) G \hat{r}.$$

Because $\hat{z}$ is a constant Cartesian vector and $\hat{z} \cdot \hat{r} = \cos \theta$,

$$\psi \equiv \vec{\nabla} \cdot (\hat{z} G) = \hat{z} \cdot \vec{\nabla} G = -\left(ik + \frac{1}{r} \right) G \cos \theta.$$

Write $\psi = g(r) \cos \theta$, where

$$g(r) = -\frac{1}{4\pi} \left(\frac{ik}{r} + \frac{1}{r^2} \right) e^{-ikr}.$$

Radiation from a Hertzian Dipole (cont.)

The radial differentiation is

$$\begin{aligned} g'(r) &= -\frac{e^{-ikr}}{4\pi} \left[-\frac{ik}{r^2} - \frac{2}{r^3} + \left(\frac{ik}{r} + \frac{1}{r^2} \right) (-ik) \right] \\ &= -\frac{e^{-ikr}}{4\pi} \left[\frac{k^2}{r} - \frac{2ik}{r^2} - \frac{2}{r^3} \right] \\ &= G \left(-k^2 + \frac{2ik}{r} + \frac{2}{r^2} \right), \end{aligned}$$

whereas the polar differentiation is

$$\frac{1}{r} \partial_\theta \psi = -\frac{g(r)}{r} \sin \theta = \frac{1}{r} \left(ik + \frac{1}{r} \right) G \sin \theta.$$

Thus no derivative step is implicit:

$$\begin{aligned} \vec{\nabla} [\vec{\nabla} \cdot (\hat{z}G)] &= G \left(-k^2 + \frac{2ik}{r} + \frac{2}{r^2} \right) \cos \theta \hat{r} \\ &\quad + \frac{G}{r} \left(ik + \frac{1}{r} \right) \sin \theta \hat{\theta}. \end{aligned} \tag{3.3.4}$$

Radiation from a Hertzian Dipole (cont.)

The cross product needed for the magnetic field is

$$\begin{aligned}\hat{z} \times \vec{\nabla} G &= (\cos \theta \hat{r} - \sin \theta \hat{\theta}) \times \left[- \left(ik + \frac{1}{r} \right) G \hat{r} \right] \\ &= - \left(ik + \frac{1}{r} \right) G \sin \theta \hat{\phi},\end{aligned}$$

where $\hat{\theta} \times \hat{r} = -\hat{\phi}$.

For the electric field, resolve $\hat{z}G = G(\cos \theta \hat{r} - \sin \theta \hat{\theta})$. The radial part inside the operator is

$$\begin{aligned}\left[\hat{z}G + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \cdot (\hat{z}G) \right]_{\hat{r}} &= G \cos \theta \left[1 - 1 + \frac{2i}{kr} + \frac{2}{k^2 r^2} \right] \\ &= G \cos \theta \left(\frac{2i}{kr} + \frac{2}{k^2 r^2} \right).\end{aligned}$$

Multiplication by $-ik\eta l \, dl$ gives

$$E_r = \frac{2\eta l \, dl}{r} \left(1 + \frac{1}{ikr} \right) G \cos \theta.$$

Radiation from a Hertzian Dipole (cont.)

Similarly, the polar part is

$$\begin{aligned}\left[\hat{z}G + \frac{1}{k^2}\vec{\nabla}\vec{\nabla} \cdot (\hat{z}G)\right]_{\hat{\theta}} &= G \sin \theta \left[-1 + \frac{1}{k^2 r} \left(ik + \frac{1}{r}\right)\right] \\ &= G \sin \theta \left(-1 + \frac{i}{kr} + \frac{1}{k^2 r^2}\right),\end{aligned}$$

and hence

$$E_{\theta} = ik\eta l \, dl \left(1 + \frac{1}{ikr} - \frac{1}{k^2 r^2}\right) G \sin \theta.$$

Combining the components, the exact fields away from the point source are

$$\begin{aligned}\vec{E} &= \frac{2\eta l \, dl}{r} \left(1 + \frac{1}{ikr}\right) \cos \theta \, G \, \hat{r} \\ &\quad + ik\eta l \, dl \left(1 + \frac{1}{ikr} - \frac{1}{k^2 r^2}\right) \sin \theta \, G \, \hat{\theta},\end{aligned}\tag{3.3.5}$$

$$\begin{aligned}\vec{H} &= -l \, dl \, \hat{z} \times \vec{\nabla} G \\ &= l \, dl \left(ik + \frac{1}{r}\right) G \sin \theta \, \hat{\phi} \\ &= ikl \, dl \left(1 + \frac{1}{ikr}\right) G \sin \theta \, \hat{\phi}.\end{aligned}\tag{3.3.6}$$

Radiation from a Hertzian Dipole (cont.)

These expressions include terms that scale as $1/r$, $1/r^2$, and $1/r^3$, corresponding to radiative, inductive, and electrostatic components, respectively. This decomposition enables classification of the field into near-field ($kr \ll 1$) and far-field ($kr \gg 1$) regions. In the near field, set $G \approx 1/(4\pi r)$ and retain the highest inverse power in each component:

$$\begin{aligned}E_r &\approx \frac{2\eta l dl}{r} \frac{1}{ikr} \frac{\cos \theta}{4\pi r} = -\frac{i\eta l dl}{4\pi kr^3} 2 \cos \theta, \\E_\theta &\approx ik\eta l dl \left(-\frac{1}{k^2 r^2}\right) \frac{\sin \theta}{4\pi r} = -\frac{i\eta l dl}{4\pi kr^3} \sin \theta, \\H_\phi &\approx ikl dl \frac{1}{ikr} \frac{\sin \theta}{4\pi r} = \frac{l dl}{4\pi r^2} \sin \theta.\end{aligned}$$

Thus

$$\vec{E} \approx -i \frac{\eta l dl}{4\pi kr^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}), \quad (3.3.7)$$

$$\vec{H} \approx \frac{l dl}{4\pi r^2} \sin \theta \hat{\phi}. \quad (3.3.8)$$

Radiation from a Hertzian Dipole (cont.)

In the far-field, only the $1/r$ terms survive, and the fields simplify to:

$$\vec{E} \approx ik\eta I dl \sin \theta G \hat{\theta}, \quad (3.3.9)$$

$$\vec{H} \approx ik I dl \sin \theta G \hat{\phi}. \quad (3.3.10)$$

In this region, the fields are transverse, mutually orthogonal, and satisfy $\vec{H} = \hat{r} \times \vec{E}/\eta$. In lossless free space k, η are real, while I may be a complex peak-amplitude phasor. Since $\hat{\theta} \times \hat{\phi} = \hat{r}$, the time-averaged Poynting vector of (1.5.6) is

$$\begin{aligned} \vec{S}_{\text{avg}} &= \frac{1}{2} \Re\{\vec{E} \times \vec{H}^*\} \\ &\approx \frac{1}{2} \hat{r} (ik\eta I dl G)(ik I dl G)^* \sin^2 \theta \\ &= \hat{r} \frac{\eta}{2} |k I dl|^2 \sin^2 \theta |G|^2 \\ &= \hat{r} \frac{\eta |k I dl|^2}{32\pi^2 r^2} \sin^2 \theta. \end{aligned} \quad (3.3.11)$$

Radiation from a Hertzian Dipole (cont.)

The angular integral needed for the total radiated power is

$$\int_0^{2\pi} d\phi \int_0^\pi \sin^3 \theta d\theta = 2\pi \left[-\cos \theta + \frac{\cos^3 \theta}{3} \right]_0^\pi = \frac{8\pi}{3}.$$

Therefore

$$\begin{aligned} P_{\text{rad}} &= \int_0^{2\pi} \int_0^\pi \vec{S}_{\text{avg}} \cdot \hat{r} r^2 \sin \theta d\theta d\phi \\ &= \frac{\eta |kl dl|^2}{32\pi^2} \frac{8\pi}{3} = \frac{\eta |kl dl|^2}{12\pi}. \end{aligned} \quad (3.3.12)$$

The radiation intensity is defined as $U(\theta, \phi) = r^2 \vec{S}_{\text{avg}} \cdot \hat{r}$, which for the Hertzian dipole becomes

$$U(\theta) = \frac{\eta |kl dl|^2}{32\pi^2} \sin^2 \theta. \quad (3.3.13)$$

The normalized directive-gain pattern is the ratio of the radiation intensity in a given direction to its spherical average:

$$D(\theta) = \frac{U(\theta)}{P_{\text{rad}}/(4\pi)} = \frac{3}{2} \sin^2 \theta. \quad (3.3.14)$$

Radiation from a Hertzian Dipole (cont.)

Here, the term $P_{\text{rad}}/4\pi$ represents the *average radiated power per unit solid angle*. This shows that the Hertzian dipole radiates with maximum intensity in the plane perpendicular to its axis ($\theta = \pi/2$) and no radiation along the axis itself ($\theta = 0$ or π). Its directivity, the maximum of $D(\theta)$, is $D_0 = 3/2$. The three-dimensional directivity pattern of a Hertzian dipole is illustrated in Fig. 3.1.

From the duality principle, the far field of a magnetic dipole source can be expressed as

$$\vec{E} \approx -ikI_m dl \sin \theta G \hat{\phi}, \quad (3.3.15)$$

$$\vec{H} \approx ik \frac{I_m}{\eta} dl \sin \theta G \hat{\theta}, \quad (3.3.16)$$

where $I_m dl$ is the magnetic-current moment (units V m). Equivalent magnetic currents are mathematical sources. To relate the magnetic-current element to a physical electrically small loop, consider a thin closed loop C carrying an approximately uniform current I_ℓ . Let its dimensions be of order ℓ , with $k\ell \ll 1$, and choose the origin inside the loop. Its electric-current radiation integral is

$$\vec{N}(\hat{r}) = I_\ell \oint_C e^{ik\hat{r} \cdot \vec{r}'} d\vec{r}'.$$

Radiation from a Hertzian Dipole (cont.)

Expanding the exponential, the zeroth-order term vanishes because $\oint_C d\vec{r}' = 0$. Define the oriented area and dipole moment by

$$\vec{S} = \frac{1}{2} \oint_C \vec{r}' \times d\vec{r}', \quad \vec{m} = I_\ell \vec{S}.$$

The identity $\oint_C d(r'_i r'_j) = 0$ makes $\oint_C r'_i dr'_j$ antisymmetric, and hence

$$\oint_C (\hat{r} \cdot \vec{r}') d\vec{r}' = \vec{S} \times \hat{r}.$$

For example, contraction of $\oint_C r'_i dr'_j = \epsilon_{ijk} S_k$ with $\hat{r}_i$ gives this vector identity directly. Therefore

$$\vec{N} = ik(\vec{m} \times \hat{r}) + O(|I_\ell| k^2 \ell^3).$$

The retained term is transverse. The electric-current far-field formula then gives, with $G_r = e^{-ikr}/(4\pi r)$,

$$\vec{E} \approx -ik\eta G_r \vec{N} = k^2 \eta G_r (\vec{m} \times \hat{r}),$$

$$\vec{H} \approx \frac{1}{\eta} \hat{r} \times \vec{E} = k^2 G_r [\vec{m} - \hat{r}(\hat{r} \cdot \vec{m})].$$

Radiation from a Hertzian Dipole (cont.)

For $\vec{m} = I_\ell S \hat{z}$, these become

$$\vec{E} \approx k^2 \eta I_\ell S G_r \sin \theta \hat{\phi}, \quad \vec{H} \approx -k^2 I_\ell S G_r \sin \theta \hat{\theta}.$$

Comparison with the preceding magnetic-current fields now determines

$$I_m dl = ik\eta I_\ell S = i\omega\mu I_\ell S.$$

This is a leading dipole-order far-field equivalence for a small loop, not an exact representation of a finite loop at every distance.

Radiation from a Hertzian Dipole

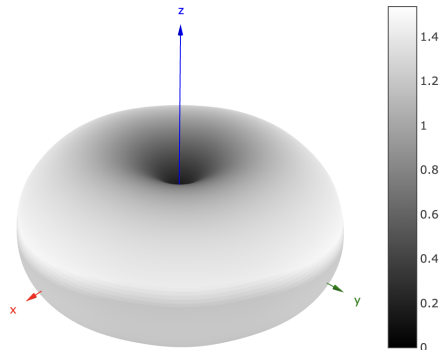


Figure 3.1: Hertzian-dipole directivity $D(\theta) = \frac{3}{2} \sin^2 \theta$: the z-axis is a null and the equatorial maximum is $D_0 = 1.5$.

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Image theory constructs a source configuration whose field satisfies the same differential equation, conductor boundary condition, and condition at infinity as the original problem; uniqueness then identifies the two fields in the region of interest. As a first example, place a charge q at $(0, 0, h)$, $h > 0$, above a grounded PEC plane $z = 0$. Put an image charge $-q$ at $(0, 0, -h)$. In $z > 0$ the proposed potential is

$$\varphi(x, y, z) = \frac{q}{4\pi\epsilon} \left[\frac{1}{\sqrt{x^2 + y^2 + (z - h)^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z + h)^2}} \right]. \quad (3.4.1)$$

At the conductor,

$$\varphi(x, y, 0) = \frac{q}{4\pi\epsilon} \left[\frac{1}{\sqrt{x^2 + y^2 + h^2}} - \frac{1}{\sqrt{x^2 + y^2 + h^2}} \right] = 0.$$

Image Theory (cont.)

The image singularity lies outside the physical half-space. Consequently, in $z > 0$

$$\nabla^2 \varphi = -\frac{q}{\epsilon} \delta(x) \delta(y) \delta(z - h),$$

and $\nabla^2 \varphi = 0$ only away from the real charge. Thus the proposed potential satisfies Poisson's equation at the source, Laplace's equation elsewhere, the grounded-PEC boundary value, and decay at infinity. These are the hypotheses that make the upper-half-space solution unique.

For time-harmonic sources the same method applies when the conductor is ideal and the image orientation is chosen to enforce the boundary condition at every point on the plane. A PEC requires $\hat{n} \times \vec{E} = 0$; a PMC requires $\hat{n} \times \vec{H} = 0$.

The image rules must be checked with the exact dipole field, not merely with the far-field approximation (3.3.9). For a current moment $I d\vec{l}$, rotationally generalizing the exact components derived above gives

$$\vec{E}(\hat{p}, \vec{R}) = \alpha(R) \hat{p} + \beta(R) (\hat{p} \cdot \hat{R}) \hat{R}, \quad (3.4.2)$$

Image Theory (cont.)

where $\vec{R} = \vec{r} - \vec{r}'$ is the source-to-field vector, $R = |\vec{R}|$, $\hat{R} = \vec{R}/R$, $G_R = e^{-ikR}/(4\pi R)$, and

$$\begin{aligned}\alpha(R) &= \eta l \, dl \, G_R \left(-ik - \frac{1}{R} + \frac{i}{kR^2} \right), \\ \beta(R) &= \eta l \, dl \, G_R \left(ik + \frac{3}{R} - \frac{3i}{kR^2} \right).\end{aligned}\tag{3.4.3}$$

To see these coefficients directly, write $\hat{p} = \cos \theta \, \hat{R} - \sin \theta \, \hat{\theta}$. Equation (3.4.2) then has components

$$E_R = (\alpha + \beta) \cos \theta, \quad E_\theta = -\alpha \sin \theta.$$

Matching them to the exact E_r, E_θ above gives

$$\begin{aligned}-\alpha &= ik\eta l \, dl \, G_R \left(1 + \frac{1}{ikR} - \frac{1}{k^2 R^2} \right), \\ \alpha + \beta &= \frac{2\eta l \, dl}{R} G_R \left(1 + \frac{1}{ikR} \right),\end{aligned}$$

whose solution is precisely (3.4.3).

Image Theory (cont.)

Let the real source be at $(0, 0, h)$, its image at $(0, 0, -h)$, and the field point be $(x, y, 0)$. The two source-to-field vectors then share the magnitude $R = \sqrt{x^2 + y^2 + h^2}$; define the common in-plane vector

$$\vec{t} = \frac{x\hat{x} + y\hat{y}}{R}.$$

The two source-to-field unit vectors and Green functions obey

$$\hat{R}_+ = \vec{t} - \frac{h}{R}\hat{z}, \quad \hat{R}_- = \vec{t} + \frac{h}{R}\hat{z}, \quad G_{R_+} = G_{R_-};$$

therefore the same scalars $\alpha(R)$ and $\beta(R)$ apply to both fields.

For a vertical electric dipole, choose $\hat{p}_+ = \hat{p}_- = \hat{z}$. Taking the in-plane part of the sum in (3.4.2),

$$(\vec{E}_+ + \vec{E}_-)_{\parallel} = \alpha(\vec{0} + \vec{0}) + \beta \left[\left(-\frac{h}{R}\right) \vec{t} + \left(\frac{h}{R}\right) \vec{t} \right] = \vec{0}.$$

Thus a vertical electric dipole has a same-oriented image above a PEC. For any horizontal orientation $\hat{p} \cdot \hat{z} = 0$, choose $\hat{p}_+ = \hat{p}$ and $\hat{p}_- = -\hat{p}$. Since

$$\hat{p}_+ \cdot \hat{R}_+ = \hat{p} \cdot \vec{t}, \quad \hat{p}_- \cdot \hat{R}_- = -\hat{p} \cdot \vec{t},$$

Image Theory (cont.)

the tangential sum is

$$(\vec{E}_+ + \vec{E}_-)_{\parallel} = \alpha(\hat{p} - \hat{p}) + \beta [(\hat{p} \cdot \vec{t})\vec{t} - (\hat{p} \cdot \vec{t})\vec{t}] = \vec{0}.$$

Hence a horizontal electric dipole has an oppositely oriented image. These cancellations use all $1/R$, $1/R^2$, and $1/R^3$ terms and hold at every nonsource point of the plane. The exact PEC boundary condition and the outgoing radiation condition now justify the uniqueness argument in the upper half-space.

Before using duality, check the electric-dipole image for a PMC plane independently. The exact magnetic field follows from $\vec{A} = \mu I d\ell G(R)\hat{p}$:

$$\vec{H} = I d\ell \vec{\nabla} G \times \hat{p} = C(R)\hat{p} \times \hat{R}, \quad C(R) = I d\ell \left(ik + \frac{1}{R} \right) G(R).$$

On $z = 0$, write the equal-distance displacement directions from a source at $+h$ and its image at $-h$ as $\hat{R}_+ = \vec{t} - (h/R)\hat{z}$ and $\hat{R}_- = \vec{t} + (h/R)\hat{z}$, with $\vec{t}$ tangent to the plane. For a perpendicular electric dipole choose opposite image orientation:

$$\hat{z} \times \hat{R}_+ - \hat{z} \times \hat{R}_- = 0.$$

Image Theory (cont.)

For a tangential electric dipole choose the same image orientation:

$$\hat{\mathbf{p}} \times \hat{\mathbf{R}}_+ + \hat{\mathbf{p}} \times \hat{\mathbf{R}}_- = 2\hat{\mathbf{p}} \times \vec{\mathbf{t}},$$

which is normal to the plane. Both choices give $\hat{\mathbf{z}} \times \vec{\mathbf{H}} = 0$, the PMC condition. Thus the electric/PMC signs are opposite to the already derived electric/PEC signs.

The duality map derived above is $\vec{\mathbf{E}}_d = -\eta \vec{\mathbf{H}}$, $\vec{\mathbf{H}}_d = \vec{\mathbf{E}}/\eta$, $\vec{\mathbf{M}}_d = \eta \vec{\mathbf{J}}$. It exchanges PEC and PMC boundaries and turns electric sources into magnetic sources without changing a relative image sign. Applying it to each of the two electric-source rows supplies the two magnetic-source rows.

For completeness, planar image uniqueness does not require treating an infinite plane as a bounded obstacle. Reflect a homogeneous PEC difference field across the plane with tangential $\vec{\mathbf{E}}$ odd and normal $\vec{\mathbf{E}}$ even. The tangential components vanish on the plane, and Gauss' law gives $\partial_z E_z = 0$ there, so the reflected Cartesian components solve the all-space homogeneous Helmholtz equation without a surface delta source. They are regular and outgoing; the all-space result in Section 3.1.3 makes them zero. For PMC

Image Theory (cont.)

use the same argument with $\vec{H}$. This justifies the image solution in the appropriate half-space. The four orientation rules are therefore

	$\perp$	$\parallel$
PEC, electric	+	-
PEC, magnetic	-	+
PMC, electric	-	+
PMC, magnetic	+	-

Here + means the same image orientation and - the opposite orientation; $\perp$ and $\parallel$ refer to the plane. Figure 3.2 summarizes these four sign rules.

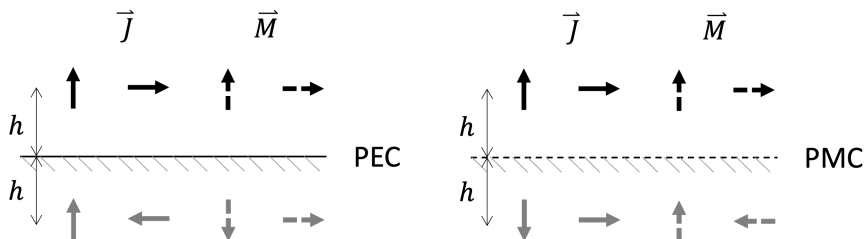


Figure 3.2: Image orientations for electric-current moments $\vec{J}$ and magnetic-current moments $\vec{M}$ above PEC and PMC planes. Black arrows are the real moments, gray arrows are their images, and dashed arrows denote magnetic moments; the orientations agree with the sign table in the text.

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Radiation from an Aperture

When an electromagnetic wave passes through an opening in a conducting screen, the field in the observation half-space can be represented by equivalent currents on the aperture.

Let the aperture lie in $z = 0$, let the desired radiated field occupy $z > 0$, and choose $\hat{n} = +\hat{z}$. With this orientation, Love's equivalent currents are

$$\vec{J}_s = \hat{z} \times \vec{H}_a, \quad \vec{M}_s = -\hat{z} \times \vec{E}_a, \quad (3.5.1)$$

where $\vec{E}_a, \vec{H}_a$ are the tangential aperture fields. Reversing the chosen normal (and the retained half-space) reverses both current signs. To simplify the analysis, consider the transformations in Fig. 3.3. From the surface equivalence principle (1.5.33a), the PEC in the original problem is first replaced by surface currents on the z -plane. Since $\vec{E}_t$, the tangential component of the electric field, vanishes on the PEC, the corresponding equivalent magnetic surface current $\vec{M}_s$ is also zero, whereas $\vec{J}_s = \hat{z} \times \vec{H}$ need not vanish at this stage. In the second step, apply the PEC replacement in (1.5.33b) to the suppressed half-space $z < 0$ (region V_1 in Fig. 3.3), while retaining the desired field in $z > 0$. With this PEC retained, the impressed equivalent tangential electric-current sheet may be omitted on the entire $z = 0$ plane, leaving only $\vec{M}_s = -\hat{z} \times \vec{E}_a$ on the aperture. This does not imply that the induced PEC surface current vanishes. Finally,

Radiation from an Aperture (cont.)

the image theory removes the PEC and doubles the equivalent magnetic surface current: since $\vec{M}_s = -\hat{z} \times \vec{E}_a$ is tangential to the $z = 0$ plane, it is a *horizontal* magnetic current, whose image in a PEC has the *same* orientation (Fig. 3.2). The source and its image therefore reinforce rather than cancel, so removing the conductor leaves a single magnetic current of twice the strength radiating into free space:

$$\vec{M}_s = -2\hat{z} \times \vec{E}_a. \quad (3.5.2)$$

Substituting $\vec{M}_s = -2\hat{z} \times \vec{E}_a$ into the magnetic-current term of (3.2.12), with $-\hat{r} \times \vec{M}_s = 2\hat{r} \times (\hat{z} \times \vec{E}_a)$ and the integral taken over the aperture area S_A , gives

$$\vec{E} \approx -ik \frac{e^{-ikr}}{4\pi r} \int_{S_A} 2\hat{r} \times (\hat{z} \times \vec{E}_a) e^{ik\hat{r} \cdot \vec{r}'} ds',$$

so the factor of 2 is absorbed into the prefactor, replacing $1/(4\pi r)$ by $1/(2\pi r)$ in (3.5.3).

Radiation from an Aperture

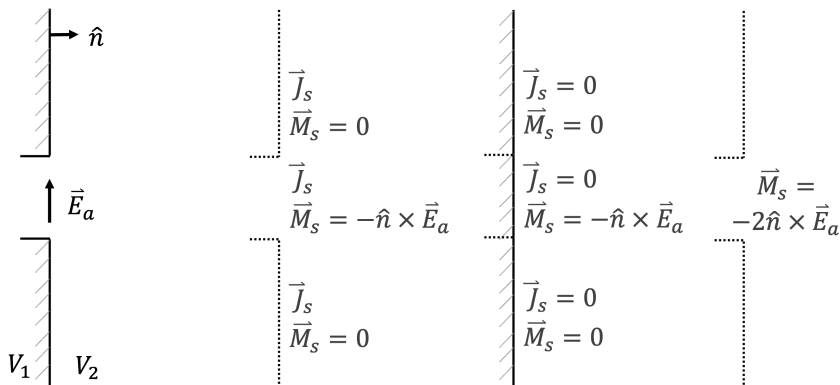


Figure 3.3: Aperture equivalence sequence for the retained half-space: replace the screen by equivalent currents, restore a PEC so only $\vec{M}_s = -\hat{n} \times \vec{E}_a$ remains on the aperture, and then remove the PEC by doubling the same-oriented magnetic current.

Radiation from an Aperture

The far-field electric field follows from (3.2.12) applied to this surface magnetic current:

$$\vec{E} \approx -ik \frac{e^{-ikr}}{2\pi r} \int_{S_A} \hat{r} \times (\hat{z} \times \vec{E}_a) e^{ik\hat{r} \cdot \vec{r}'} ds'. \quad (3.5.3)$$

Consider a rectangular aperture of dimensions a and b along the x - and y -axes. As an analytically tractable aperture-field model, prescribe $\vec{E}_a = E_0 \hat{y}$ uniformly over the opening and zero outside. A normally incident uniform plane wave motivates this model, but illumination alone does not make the exact aperture field uniform: edge diffraction and coupling generally modify it. Under the stated uniform-field approximation, (3.5.3) can be evaluated explicitly. To reduce the integrand, apply the BAC–CAB identity $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$ with $\vec{a} = \hat{r}$, $\vec{b} = \hat{z}$, and $\vec{c} = \vec{E}_a = E_0 \hat{y}$:

$$\hat{r} \times (\hat{z} \times \vec{E}_a) = E_0 [\hat{z}(\hat{r} \cdot \hat{y}) - \hat{y}(\hat{r} \cdot \hat{z})].$$

Resolving the Cartesian unit vectors into the spherical basis,

$$\begin{aligned} \hat{y} &= \sin \theta \sin \phi \hat{r} + \cos \theta \sin \phi \hat{\theta} + \cos \phi \hat{\phi}, \\ \hat{z} &= \cos \theta \hat{r} - \sin \theta \hat{\theta}, \end{aligned}$$

Radiation from an Aperture (cont.)

so that the two projections are $\hat{r} \cdot \hat{y} = \sin \theta \sin \phi$ and $\hat{r} \cdot \hat{z} = \cos \theta$. Substituting these,

$$\begin{aligned} \hat{r} \times (\hat{z} \times \vec{E}_a) = E_0 \bigg[& \sin \theta \sin \phi \left(\cos \theta \hat{r} - \sin \theta \hat{\theta} \right) \\ & - \cos \theta \left(\sin \theta \sin \phi \hat{r} + \cos \theta \sin \phi \hat{\theta} \right. \\ & \left. + \cos \phi \hat{\phi} \right) \bigg]. \end{aligned}$$

The two $\hat{r}$ contributions, $\sin \theta \cos \theta \sin \phi$ and $-\cos \theta \sin \theta \sin \phi$, cancel, confirming that the far field is transverse. Collecting the $\hat{\theta}$ terms, $-\sin^2 \theta \sin \phi - \cos^2 \theta \sin \phi = -\sin \phi$, and the $\hat{\phi}$ term, $-\cos \theta \cos \phi$, leaves

$$\hat{r} \times (\hat{z} \times \vec{E}_a) = -E_0 \left(\sin \phi \hat{\theta} + \cos \theta \cos \phi \hat{\phi} \right).$$

Define the two direction cosines $u = \sin \theta \cos \phi$ and $v = \sin \theta \sin \phi$. The leading factor $-ik$ in (3.5.3) flips this sign, giving:

$$\begin{aligned} \vec{E} \approx ikE_0 \frac{e^{-ikr}}{2\pi r} \int_{-a/2}^{a/2} \int_{-b/2}^{b/2} \begin{pmatrix} \sin \phi \hat{\theta} \\ + \cos \theta \cos \phi \hat{\phi} \end{pmatrix} \\ e^{ik(x'u+y'v)} dy' dx', \end{aligned} \tag{3.5.4}$$

Radiation from an Aperture (cont.)

The polarization vector $\sin \phi \hat{\theta} + \cos \theta \cos \phi \hat{\phi}$ is constant in the integration variables x' and y' , so it factors out of the surface integral, which then separates into a product of two one-dimensional integrals. Each is elementary:

$$\begin{aligned}\int_{-a/2}^{a/2} e^{ikux'} dx' &= \left. \frac{e^{ikux'}}{iku} \right|_{-a/2}^{a/2} = \frac{e^{ikua/2} - e^{-ikua/2}}{iku} \\ &= \frac{2 \sin\left(\frac{kua}{2}\right)}{ku} = a \operatorname{sinc}\left(\frac{ka}{2} u\right),\end{aligned}$$

and, in the same way, $\int_{-b/2}^{b/2} e^{ikvy'} dy' = b \operatorname{sinc}(kbv/2)$. Here $\operatorname{sinc} x = \sin x/x$, with its continuous limiting value $\operatorname{sinc} 0 = 1$. Multiplying the two results gives

$$\begin{aligned}\vec{E} &\approx ikE_0 ab \frac{e^{-ikr}}{2\pi r} \begin{pmatrix} \sin \phi \hat{\theta} \\ + \cos \theta \cos \phi \hat{\phi} \end{pmatrix} \\ &\quad \operatorname{sinc}\left(\frac{ka}{2} u\right) \operatorname{sinc}\left(\frac{kb}{2} v\right),\end{aligned}\tag{3.5.5}$$

Radiation from an Aperture (cont.)

The two sinc factors are the rectangular-aperture transform, but they are not the entire vector pattern. The polarization (obliquity) factor has squared magnitude

$$|\sin \phi \hat{\theta} + \cos \theta \cos \phi \hat{\phi}|^2 = \sin^2 \phi + \cos^2 \theta \cos^2 \phi = 1 - u^2.$$

Thus, apart from angle-independent constants, the complete power pattern in the retained observation half-space $0 \leq \theta \leq \pi/2$ is

$$(1 - u^2) \operatorname{sinc}^2\left(\frac{ka}{2}u\right) \operatorname{sinc}^2\left(\frac{kb}{2}v\right).$$

Further Reading

The exposition of radiation solutions within this chapter primarily draws upon Sheng's works [24, 25]. For comprehensive details regarding notations employing dyadic Green functions, the reader is directed to pertinent chapters in Pathak's text [23] and Tai's authoritative book on this topic [28]. The Stratton–Chu formulation is extensively presented in Ishimaru's treatise [12]. The application of the equivalence principle to aperture radiation is also demonstrated in S. K. Jeng's paper [14], which shows how the aperture field due to a plane wave incident on a cavity-backed slit can be obtained, and how the corresponding radiation from the aperture, modeled as a magnetic current, may be computed using the same method described in this section.

Certain advanced topics, including Liénard–Wiechert potentials and radiation from moving charges, fall outside the scope of this discussion. For these subjects, readers are encouraged to consult canonical graduate-level electrodynamics textbooks such as that by Zangwill [31], or specialized volumes on radiation, such as Smith [26].

- 1 Verify (3.1.27) and (3.1.28) by applying the duality transform to (3.1.25) and (3.1.26).
- 2 Following Subsection 3.1.1, first consider magnetic sources with electric sources absent. The electric flux density may be written using the electric vector potential

$$\vec{D} = -\vec{\nabla} \times \vec{F}.$$

Define the magnetic field in terms of the electric vector potential and magnetic scalar potential by

$$\vec{H} = -\vec{\nabla} \vartheta - i\omega \vec{F}.$$

Assuming that the Lorenz condition holds for electric vector potentials,

$$\vec{\nabla} \cdot \vec{F} = -i\omega\mu\epsilon\vartheta.$$

Then use superposition of the electric-source potentials $(\vec{A}, \varphi)$ and magnetic-source potentials $(\vec{F}, \vartheta)$ to show that, when both source types are present, the fields have the general forms

$$\vec{E} = -i\omega\vec{A} + \frac{1}{i\omega\mu\epsilon}\vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \frac{1}{\epsilon}\vec{\nabla} \times \vec{F},$$

Problems (cont.)

$$\vec{H} = -i\omega\vec{F} + \frac{1}{i\omega\mu\epsilon}\vec{\nabla}(\vec{\nabla}\cdot\vec{F}) + \frac{1}{\mu}\vec{\nabla}\times\vec{A}.$$

- 3 For an observation point strictly outside the source support, show that (3.1.31) is equal to

$$\begin{aligned} -ik\eta \int_{V'} \left[\left(\bar{\bar{I}} + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \right) G(\vec{r}; \vec{r}') \right] \cdot \vec{J}(\vec{r}') dV' \\ = -ik\eta \int_{V'} \bar{\bar{G}}(\vec{r}; \vec{r}') \cdot \vec{J}(\vec{r}') dV', \end{aligned}$$

where

$$\bar{\bar{G}}(\vec{r}; \vec{r}') = \left[\left(\bar{\bar{I}} + \frac{1}{k^2} \vec{\nabla} \vec{\nabla} \right) G(\vec{r}; \vec{r}') \right]$$

is the conventional electric *dyadic Green function*.

- 4 Complete the intermediate steps in (3.3.3)–(3.3.6).

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